

Q-Research and list two real-world scenarios (such as public WiFi login or mobile data authentication) where WAN authentication protocols like PAP, CHAP, or MS-CHAP are used, and explain why authentication is critical in each case.

Here are **two real-world scenarios** where PPP authentication protocols such as PAP, CHAP, and MS-CHAP are relevant. One important correction: **ordinary public Wi-Fi captive-portal login and modern mobile-data authentication do not normally use PAP/CHAP directly**; modern cellular networks use mechanisms such as AKA. PAP/CHAP are more directly associated with PPP-based access such as broadband and VPN/remote access.

1. ISP Broadband / PPPoE Internet Connection

Scenario:

When a customer connects to an ISP using a broadband connection that uses **PPPoE (PPP over Ethernet)**, the customer's router establishes a PPP session with the ISP's access concentrator. PPPoE was specifically designed to provide per-user access control and billing.

During the PPP authentication phase, protocols such as **PAP or CHAP** can be used to verify the customer's credentials. CHAP uses a challenge-and-response mechanism rather than sending the secret directly across the link.

Why authentication is critical:

- Prevents unauthorized users from using someone else's ISP account.
- Allows the ISP to associate network access with a specific customer.
- Helps enforce service authorization and billing.
- Prevents an attacker from simply connecting to the access network and pretending to be a legitimate subscriber.

Simple example:

Customer Router → ISP Access Network → ISP Authentication Server

The router proves: "**I am an authorized customer.**"

2. Remote-Access VPN / Corporate Network

Scenario:

An employee working remotely may connect to a company's VPN gateway. PPP-based VPN systems can use authentication mechanisms including **MS-CHAP/MS-CHAPv2**. Microsoft

documentation describes MS-CHAP as being designed for authenticating remote Windows workstations, and Microsoft documents its use with PPP and VPN-related network access.

MS-CHAPv2 extends CHAP and provides **mutual authentication**, meaning the authentication process can verify both sides rather than simply accepting a client's credentials.

Why authentication is critical:

- Prevents unauthorized people from entering the company's private network.
- Protects access to internal servers, files, applications, and resources.
- Helps ensure that a person connecting remotely is actually an authorized user.
- Reduces the risk of an attacker impersonating a legitimate employee.

Simple example:

Employee Laptop → Internet → VPN Gateway → Corporate Network

Before access is granted:

Laptop ⇌ Authentication ⇌ VPN Gateway

If authentication fails, the VPN connection should not provide normal network access.